

Chapter 15

Production of Aluminum Powders[☆]

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INTRODUCTION

Aluminum and aluminum alloy powders are manufactured almost exclusively by gas atomization. Compressed air is used as the atomizing gas for most applications. Inert gases (nitrogen, argon, and helium) have found application in special cases. Currently, the annual world production of aluminum powders amounts to 100,000 metric tons [1] while the productive capacity is estimated to be approximately twice that [2]. A large part of this capacity is in North America and Russia. North American metal shipments in short tons in 1995–2015 (Fig. 15.1) show the tendency of the dynamics of aluminum powder production up to 2013, and stabilization in 2014–15, at the level of 40,000 tons [3].

The ranges of aluminum powder applications are of great variety. They include aluminothermic reduction in metallurgy, chemical processes, explosives, rocket fuels, pyrotechnics, thermite welding, additives for gas generation in aero-concrete production, pharmaceuticals, and pigments for paints and inks.

These are the following basic kinds of powder products from aluminum:

- Powders: particles with a size from a few microns to 1.0 mm (usually, not >0.5 mm) of spherical, oval, drop-like, irregular, fragmental, or dendritic shape.
- Flakes: scaly (lamellar) particles with a thickness smaller than 1.0 μm and linear sizes 50–200 μm . Particles are usually coated with a thin layer of stearin, aluminum stearate, paraffin, or other fats, the content of which is 0.2%–4.0%.
- Pastes: compositions of flake particles (65%–75%) and a liquid phase (water, hydrocarbons).
- Granules (grit): particles with a size larger than 0.5–1.0 mm of spherical, oval, drop-like, disc-shaped, needle, or irregular shape.

Known methods of aluminum powder production can be divided into two basic groups:

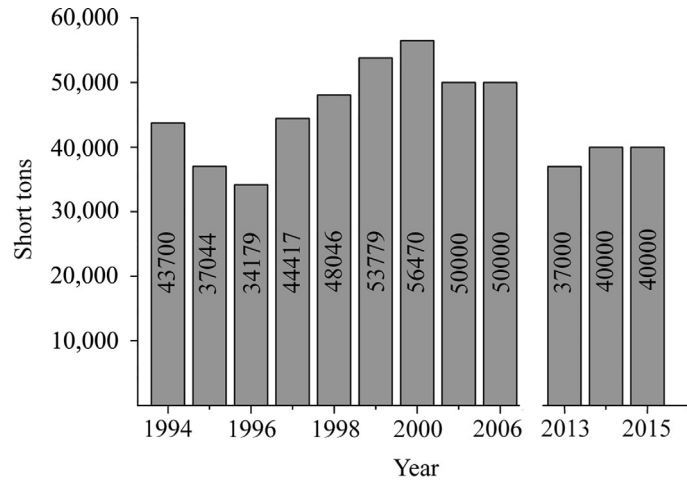
- Physicochemical methods: electrolytic sedimentation from solutions or melts; reduction (or thermal decomposition) of aluminum compounds; and evaporation and condensation of the metal from a vapor phase.
- Physicomechanical methods, that is, without basic changes of chemical and phase structure of the initial material: mechanical crushing of aluminum, fragile alloys of chips, foil; atomization of melts.

Along with the main commercial gas-atomization technique, the following methods for aluminum powder manufacture are known:

- *Crushing* of fragile aluminum alloys (Al–Ni, Al–Mg, Al–Si) in ball mills or other crushing devices (particle shape is fragmental).

[☆] Expanded and updated by O.D. Neikov.

FIG. 15.1 North American aluminum powder shipments in short tons in 1994–2015. From Williams B. Powder metallurgy—a global market review. In: International powder metallurgy directory & yearbook, 12th ed.; 2006/2007. p. 5–16, MPIF2016(5) www.mpiif.org.



- *Dry grinding* of aluminum powder or foil in the presence of surfactants in a protective gas atmosphere with monitored content of oxygen (production of pigment, pyrotechnic, and clumpy powders).
- *Grinding of aluminum particles* in a liquid phase (hydrocarbons, water) containing surfactants (stearin) and other additives (production of pastes). Pastes can also be manufactured by mixing the powder with hydrocarbons or aqueous solutions with special additives.
- *Granulation*: free casting of the melt into a cooling medium through ground apertures of the melting pot (crucible); the size of the granules is of 4–20 mm (the shape is disk-shaped).
- *Centrifugal atomization* of the melt through lateral apertures of the punched walls of a rotating glass, from a flat disk surface, edges of a cone glass, etc. The molten metal forms droplets that solidify in the cooling gas medium or in water. Plants for centrifugal atomization of the melt are mainly used to manufacture granules with a grain size of >0.5 mm in size or coarse fractions of powders larger than $100\mu\text{m}$ in size. Centrifugal atomization is usually carried out through apertures of the rotating punched vessel (glass). Designs of plants with centrifugal atomization are used in Russia and Ukraine [4].
- *Disintegration of solid metal* at hot brittleness temperature; the size of granules is 5–20 mm, and their shape is irregular.
- *Lumping of fine foil scrap* into round granules of 3–6 mm in size.

The characteristics of aluminum powders produced by various methods are given in Table 15.1.

GAS ATOMIZATION OF ALUMINUM POWDER

Atomization of melts by compressed gases or liquids provides for the manufacture of powders of aluminum and its alloys in a wide range of sizes from $1\text{--}2\mu\text{m}$ to $1\text{--}2\text{mm}$ with particle shape varying from spherical to drop-like to irregular. Uniformity of composition and fine microstructure of powder particles are achieved by atomization of aluminum alloys due to a high cooling rate of $10^4\text{--}10^6\text{K/s}$ [5–7].

The greatest development and industrial application for producing metal powders by melt atomization by compressed gases use the following basic types of plants:

- Vertically upward aspirating plants, such as operated by Alcoa [2], allow the production of powders with the widest range of particle sizes.
- Horizontal aspirating installations, where the melt jet and atomizing torch are directed horizontally or obliquely to the horizon. The ejection feed of the melt for atomization of aluminum, magnesium, and their alloys is used at aluminum mills in Russia while a feed under pressure was utilized at the Kluchevskoy ferroalloy mill in Russia and a siphon feed was used at the Ekkart-Werke in Germany [4].
- Vertical installations, where a free flow stream of the liquid melt is atomized into droplets by impingement of high-pressure jets of gas or oil. They are used mainly to produce powders of heavy metals and alloys (iron, copper, complex alloys based on iron). In some cases, plants of such type are also used for producing light metals and alloys, including aluminum [4].

TABLE 15.1 Characteristics of Aluminum Powders [4]

Type of Powder, Method of Production	Shape of Particles	Mass Medium Particle Size (μm)	Density (g/cm^3)		Al_2O_3 (wt%)	Al Content (wt%)
			Apparent	Tap Density		
Powders produced by:						
Air	Oval with rough surface shape, irregular	50–200	0.73–1.24	1.15–1.25	0.1–1.0	99.5
Inert gas atomization	Spheroidal	45–100	0.5–1.4	0.90–1.70	0.1–1.7	99.5
Ultrafine powders produced by condensation from gas medium	Spheroidal	0.2–0.5	...	...	...	>99
Powders produced by grinding fragile alloys of Al–Mg, Al–Ni, Al–Si systems	Fission-fragment	40–2000	...	...	...	70–90
Flakes produced by dry grinding:						
Fine	Flaky	45	0.27–0.41	0.41–0.68	2–21	96–98
Coarse	Plate-shaped	150–1170	0.22	0.36	1	>99.5
Cloddy (for SAP)	Lump of flaky particles	210	1.08–1.21	1.35–1.49	6–14	The rest
Granules produced by melt atomization	Oval, tear-shaped, irregular, acicular	500–5000	1.8–1.5	1.6–1.8	...	87–99.5
Granules produced by free melt casting into the water or cooled surface	Disc-shaped	500–20,000	1.5–1.7	...	...	87–99.9
Paste produced by grinding in the fluid of liquid hydrocarbon	Flaky	45	...	...	2–20	...

Numerous other variants of constructive design of the basic types of plants for atomization of molten metals and alloys listed above are known.

A scheme of the Alcoa process based on the vertically upward aspirating principle is shown in Fig. 15.2 [8]. The molten metal is supplied from the holding or melting furnace to the atomizing tundish. The liquid is drawn from the tundish through a delivery tube into the atomizing nozzle. This is attained owing to the suction (aspiration) effect arising at the nozzle end by the flow of the high-pressure atomizing gas in the nozzle. Meeting with the high velocity gas jet, the liquid metal is broken up into droplets that are carried along by the gas flow and quenched by the gaseous cooling atmosphere in the chiller chamber. The particles formed together with the cooling air pass into the collection system that consists of two sets of cyclones. The powder collected in the cyclones is then transported in an inert gas atmosphere to the screens and packing station under inert gas.

However, this process requires careful adjustment and maintenance. An expected stopping or decreasing of the delivery of the cooling air in the chiller chamber causes an increase of the airborne dust concentration of aluminum powder over the level, which forms an explosive mixture. Generally, the technology of producing gas-atomized aluminum powders is a potentially explosive process and special safety measures are required that considerably increase the manufacturing cost. These safety measures, such as those accomplished by Alcoa, comprise cooled air supplied for the melt atomization in the chiller chamber; the carrying out under inert gas of all powder movement, including screening for separating into size

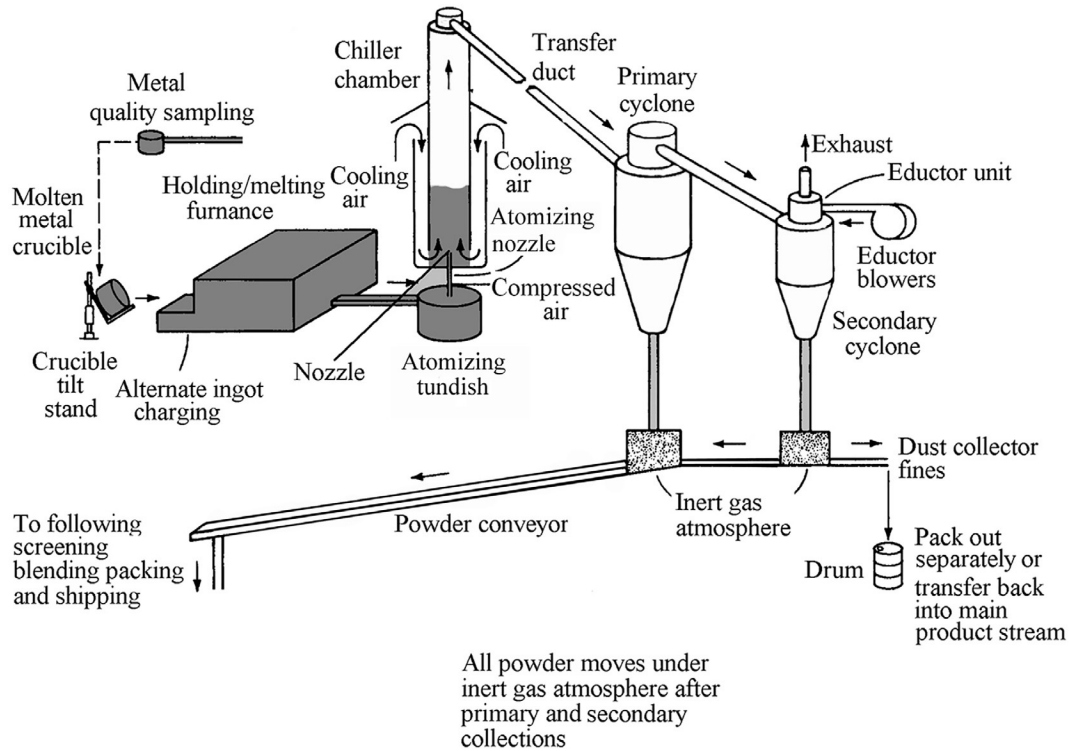


FIG. 15.2 Schematic of the Alcoa process for atomizing aluminum powder.

fractions and packing into drums; and, in addition, using sensitive systems controlling a low concentration of powder in the gas and the pressure rise.

Plants of the horizontal type have the following advantages:

- Impossibility of melt ingress into the dust-precipitating system in the case of a bath fracture.
- Relative ease of monitoring of the atomization process.
- Increase of safety due to the division of stages and devices of atomizing (melting bath, atomizer) and primary classification (atomizing chamber, cyclones).

The disadvantage of such plants is the amount of the area occupied, which is greater than for other types because a large part of the plant is in a horizontal plane.

Plants of a vertical type with top-down atomization are more compact and occupy a smaller production space. However, in this case the danger connected with an opportunity of melt inrush into the atomizing chamber is increased, the atomization process is more difficult to control, and a large chiller height is necessary for supply of solidification of the molten particle stream. The main characteristics of the explosion risks of the powders are shown in Table 15.2 [4].

The properties of typical commercial aluminum powders are given in Table 15.3.

Plants of the granulation type are lesser dangerously explosive and have a higher efficiency. However, they are not suitable for producing highly dispersed powders. The horizontal aspirating circuit of gas-atomized powder production (Fig. 15.3) most widely used in Russia includes the high compressed gas (air or inert gas) atomization systems and powder separation.

Devices with dry separation of atomization products (in cyclones, multicyclones, bag filters) and with melt atomization into a liquid medium (hydrocarbons or aqueous solutions) are used. In the latter case, higher cooling rates are achieved. A disadvantage of atomization into a liquid is the necessity of powder drying and regeneration of the coolant and received powders can be contaminated with carbon (in the case of the atomization into liquid hydrocarbons), oxygen, and hydrogen (in the case of water atomization or gas atomization into water). The high activity of aluminum powders limits the opportunities of using devices for water atomization of aluminum and its alloys.

The results of studying the influence of gas-atomization parameters on aluminum powder particle size are considered in the work [9]. The schematic diagram of an installation at the RUSAL PM manufacturing plant, where the investigations were fulfilled, is presented in Fig. 15.4. Atomization is performed by the ejection nozzle design under 3.0 MPa nitrogen pressure. Gaseous nitrogen heated by passing through a heater is supplied to the atomizer.

TABLE 15.2 Explosion Risk Characteristics

Type of Powder	Composition, Content (wt%)	Powder Particles Size (μm)	Explosion Risk Characteristics			
			LCIL (g/m^3)	p_{max} (MPa)	v_{max} (MPa/s)	τ_{in} (s)
Al	99.95Al	160 max 50 max	60 47	0.43 0.56	6.6 7.7	0.10 0.05
AD-33 (in Russian Federation classification)	1.3(Zn+Mg)-0.3Cu-0.5Si-0.14 Mn Al—balance	160 max 50 max	80 58	0.39 0.43	4.3 5.6	0.15 0.1
Al-Cr-Fe	90.5Al-8Cr-1.5Fe	63 max 50 max	101 84	0.43 0.49	2.0 4.0	0.3 0.1

LCIL: the low concentration ignition limit; p_{max} : the maximal explosion pressure; v_{max} : the maximal velocity of its growth; τ_{in} : the induction period (the time from the start of igniter to the ignition moment of the powder suspension).

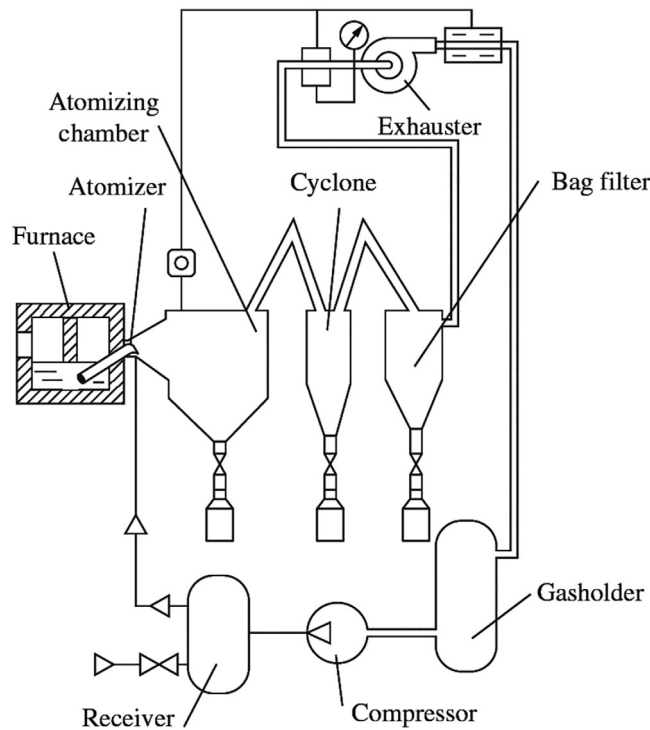


FIG. 15.3 Schematic diagram of a gas-atomization plant.

The atomizing gas is delivered through an annular orifice around the nozzle at the converging angle (Fig. 15.5). The gas flow creates the suction (ejection) effect at the tip of the nozzle that draws the molten metal from the tundish into the nozzle. In contrast of close coupled gas atomization [10] (see also Chapter 4), in present nozzle design the intake liquid metal stream is atomized by converging annular gas jet. The spraying torch is cooled by nitrogen entering from the cooling ring and the forming spray precipitated in a bottom part of the atomization chamber, which is not shown in Fig. 15.4. The amount of gas delivered by the nozzle is controlled by the heated gas tube valve.

According to data [10], the variation of atomization parameters in the following range: 1153–1253 K liquid melt temperature (T_m), 873–973 K nitrogen temperature (T_n), and 0.17–0.21 m^3/s nitrogen consumption (Q) showed that the optimum is achieved by $T_m = 1253 \text{ K}$, $T_n = 933 \text{ K}$, $Q = 0.17 \text{ m}^3/\text{s}$ condition. The optimum criterion was the maximal yield of powder particle fraction $< 10 \mu\text{m}$, which in the above condition was 21.3% by 21.2 μm median diameter.

TABLE 15.3 Properties of Typical Commercial Atomized Aluminum Powders

Type of Powder	Chemical and Physical Properties											Source
	Composition (wt%)					Apparent Density (g/cm ³)	Sieve Analysis (wt%), Sieve Size (μm)					
	Al (min)	Fe	Si	Each	Total		+250	+160	+100	+45	—45	
Atomized powders Typical	99.7	0.25 ^a	0.15 ^a	0.05 ^b	0.15 ^b	0.8–1.3	Median diameter within 5–150 μm					[2]
High-purity atomized powders Minimum Typical	99.97 99.976	0.007	0.008		0.009	0.8–1.3	Median diameter within 5–150 μm					[2]
AL-180 ^c	99.99–99.999						Inquire					AEE
Atomized aluminum powders	99.4	0.4	0.3			0.95–1.05	0	0	5–15	55–65	35–45	PM
AL-111, Coarse aluminum powder	99.7						–50+100 mech					AEE
AL-104, Spherical aluminum powder	99.9						1–3 μm					AEE
GY-FLT	99.0	0.2 ^b	0.1 ^b	0.27 ^b		0.9	1–1000 μm					Xjalp
Pigmentary aluminum powder ^d	Bal	0.5	0.4	4.06 ^e			0	0	...	0.8		RUSAL

^aFe+Si of 0.30 wt% max.^bMax.^cAluminum sputtering targets; RUSAL, Russian Federation Aluminum Company; PM, Pometon; AEE, Atlantic Equipment Engineers, Division of Micron Metals, Inc.; Xjalp, Xinji Guangyuan Metal Powder Co. Ltd.^dGas (nitrogen+6 wt% oxygen) atomized powders are ground in ball mills with greasy additives.^eIncluding 3.8% greasy additives.

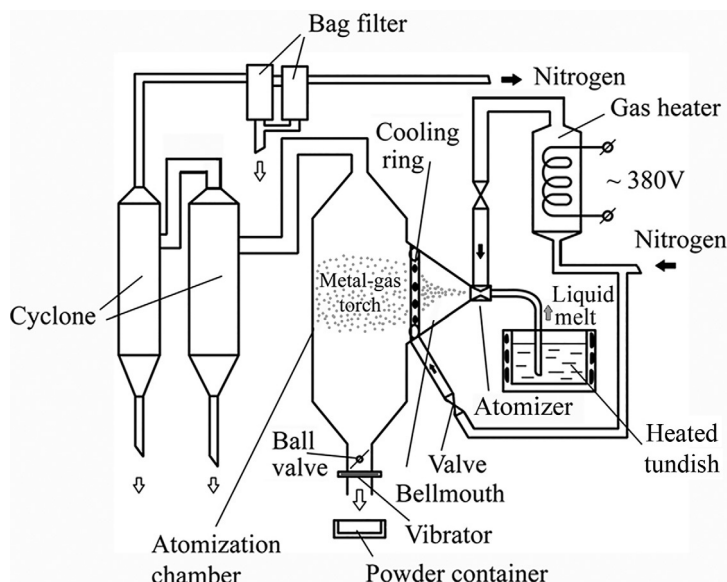


FIG. 15.4 Gas-atomization installation with ejection nozzle design at RUSAL PM manufacturing plant.

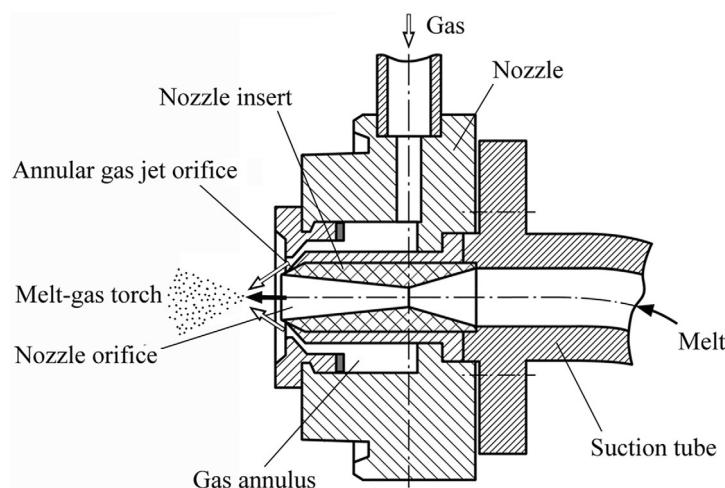


FIG. 15.5 Ejection gas atomizer.

The typical chemical analyses and physical properties of atomized aluminum powders are shown in [Table 15.3](#).

Water atomization is not used commercially for aluminum and its alloys for fear of the explosion risk of the atomization process due to rapid hydrogen emission during powder-water interaction and the high reactivity of aluminum that can cause heavy oxidation of the surface of water-atomized powders.

In contrast to the traditional opinion, recent studies have shown that water atomization under optimized conditions could be used for the production of high-quality aluminum alloy powders [11,12]. This technique is discussed in [Chapter 16](#).

PRODUCTION OF ALUMINUM GRANULES

According to the properties and application fields, granules are a transition link between powders and ingots; the precise border between powders and granules in the range of particle size from 0.1 to 1.0 mm is practically absent. In some cases, powders can be successfully replaced by granulated aluminum and its alloys, in particular in the chemical industry and in the production of steel and ferroalloys, chromium, etc.

The advantages of granules in comparison with powders are their higher profitability and the lower explosion risk, both of the production process and the product itself. This simplifies the organization of manufacture and use of such products.

The largest consumers of granules from primary and secondary aluminum are ferroalloy and steel works. The consumption of granules has increased with a corresponding decrease in the consumption of pig aluminum or powders.

In recent years, the technology for melted aluminum granulation has mainly developed in the direction of an increase of granule uniformity in size, reduction of their oxide and gas content, and a possibility of regulation by the coarseness, granule shape, and microstructure of granulated alloys. Four basic circuits of granulation technique allowing the manufacture of aluminum granules of various grain size, shape, and hardening degree are used. The basic technological schemes of granulation that are used in Russia are given in Table 15.4 and Fig. 15.6.

The scheme in Fig. 15.6A, circuit I (pouring into the water through a tundish with apertures in the bottom), is used in the enterprises of the aluminum industry and ferrous metallurgy plants. Granules of aluminum of high and technical purity as well as granulated aluminum alloys are manufactured. Mechanized lines with productivity up to 10,000 tons per year working under circuit I are used in the Volgograd Aluminum plant of the RUSAL company in Russia [4] to produce granules with a size of 5–20 mm from high-purity aluminum (for gallium cementation from aluminate solutions) and granules from secondary aluminum with high iron content (for processing pig iron and steel). In this circuit, the reduction of the granule size is achieved by vibration of the pouring bowls. The plant is completely mechanized and provides a continuous technological process from melting of the metal to discharging the commodity product classified by granule size.

Granules manufactured by centrifugal atomization of the melt into water, circuit II (Fig. 15.6B), are used in powder metallurgy. The cooling rate of granules is close to that for gas-atomized powders. The inhibition of diffusion and fixing of high-temperature structure allow a two- to three-fold increase in the level of allowable impurities in alloys without a decrease in the mechanical and corrosion properties of the material, that is, to use effectively low-grade metal. In some cases, impurities traditionally considered as harmful, such as iron, form fine intermetallic phases that provide good elevated temperature strength and other mechanical properties.

The scheme in Fig. 15.6C, circuit III, provides cooling of atomized particles in a gas atmosphere. In this process, needle-shaped (in air), drop-shaped, or spherical (in inert atmosphere or under protection of a particle surface by flux) granules are produced.

The technological circuit IV (pouring to a cooled surface, Fig. 15.6D) eliminates the contact of granules with water and the subsequent drying process. Despite the seeming simplicity in this circuit, there are serious difficulties connected with the necessity of fast heat removal from solidifying granules. As a rule, the productivity of such plants is of the order of 100–300 kg/h. Drop casting is carried out from the punched tube (funnel) radially placed over a water-cooled metal disk rotating in a horizontal plane. The granules, which have solidified on the disk, are cleaned off in the receiver, where they are led to sieving and packing. Instead of a rotating disk, a moving belt can be used. Plants of this type are used by the firms Ekkart-Werke (Germany) and Pechiney (France).

Under industrial conditions, there are plants with units that generate electromagnetic oscillations acting on the melt stream, which has allowed a reduction of granule size up to 2–10 mm [4].

In centrifugal atomization, centrifugal force breaks up the molten metal as a spray of droplets, which then solidify as powder particles. The circuits for centrifugal melt atomization (Table 15.4, circuits II and III) produce granules of various shape and size (from 0.1 up to 5 mm) with a high crystallization rate. In all cases, the cup is 100–200 mm diameter, running at moderately high speeds from 3000 to 10,000 rpm. The cooling rate estimated by calculation and experiment for the above circuits, I, II, III, and IV, is within 100–200, 100–500, 10^3 – 10^4 , and 10^2 – 10^3 K/s, respectively.

Granulated alloys of the aluminum-silicon system have been developed based on cheaper electrothermal silumin manufactured by direct reduction from ores that essentially expand the raw-material base of the aluminum industry. Cooling of the melt at a high rate in the process of granulation has allowed an increase in the content of silicon in the alloy up to 35%, thereby providing its precision properties [4].

New designs of the spinning cup have allowed the organization of the metal pouring from the cup channels as a torch that has ensured the production of a finer product and has created an opportunity to carry out “dry” atomization (the circuit VI, Fig. 15.6C). The size of the basic fraction of granules is decreased from 2.0–2.5 to 1.0–1.5 mm.

By an increase of speed rotation of the spinning cup, the centrifugal technology is capable of ensuring production granules of 0.1–0.8 mm in size. The physical properties of the granules so produced do not differ significantly from those of gas-atomized powders but they are distinguished by greater uniformity of granulometric composition and by the absence of dust fractions.

In aluminothermic processes, the application of granules with a specific surface of 200–400 cm²/g up to 1 mm in size produced by melt centrifuging are the most effective. Granules from 0.2 up to 2.0 mm in size are most effective for steel deoxidation. Coarse granules of a spherical shape of 1–5 mm in size are most suitable for the production of heat-conducting components for electric motors. Granules of spherical shape of 0.4–0.8 mm in size are used in the chemical industry in

TABLE 15.4 Characteristic of the Main Circuits of the Granulation Technique

Technological Circuit	Characteristic of Granules			Circuit Characteristic		Application Field of Granules	Device Productivity (t/h)
	Size (mm)	Shape	Particle Weight (g)	Advantage	Disadvantages		
I. Free or vibration pouring with cooling in a liquid; drying, classification	3–20	Disk-shaped, nodular	0.5–1.5	Stability of shape and size	Contact with liquid: necessity of drying; low cooling rate	Gallium cementation from aluminate solutions, ligatures, steel deoxidation, chemical industry	0.2–1.0
II. Melt centrifugal atomization with cooling in a liquid; drying; classification	0.2–2.0	Spindle-shaped, drop-like, nodular	0.002–0.05	Ease of size and structure regulation, high rate of cooling	Contact with liquid, drying of granules is necessary	Powder metallurgy, pigments, deoxidation, aluminothermy	0.6–2.5
III. Melt centrifugal atomization with cooling in gas medium; classification	0.2–2.0 thick 3–10 length	Acicular, fibrous, spindle-shaped, peteliform	0.002–0.006	No contact with liquid, no drying of granules	Large size of a collecting chamber (8 m in diameter), intensive heat mode, low rate of cooling	Aluminothermy, deoxidizers, powder metallurgy, ligatures, pigments	0.6–1.0
IV. Free or vibration pouring with cooling on a cold surface	5–40	Hemisphere, petaliform	0.5–1.5	No contact with liquid, no drying; high speed of cooling	The shape of granules are a disadvantage for compacting	Gallium cementation, chemical industry, semiproduct for grinding, ligatures, steel deoxidation, sheet rolling, pigments	0.1–2.0

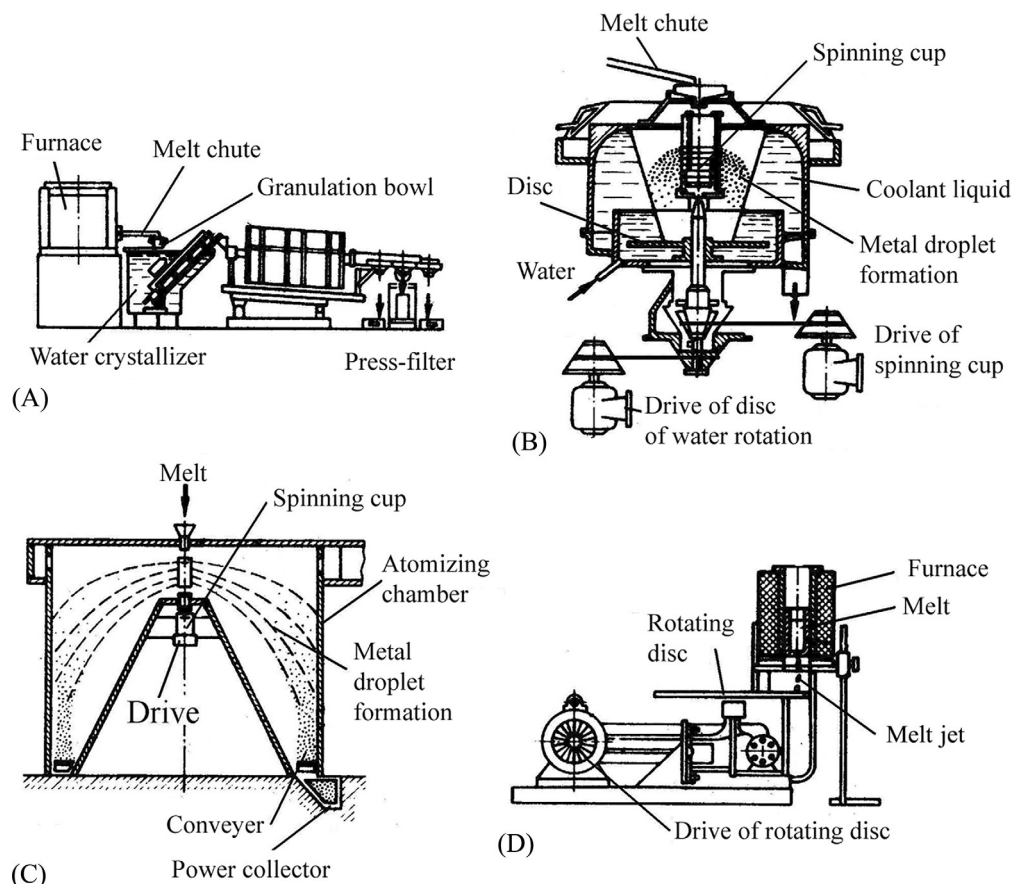


FIG. 15.6 Technique circuits of the aluminum granulation: (A) scheme of the device for production of granules by the method of free casting into a liquid (water); (B) scheme of the device for granule centrifugal atomization with cooling in a liquid; (C) scheme of the device for centrifugal atomization with cooling in gas atmosphere; and (D) scheme of the device for aluminum granulation with cooling on a cold surface.

fluidized bed devices. Granules of 7–20 mm in size from high purity aluminum are used in hydrometallurgical processes (for example, cementation of gallium from aluminate solutions).

Nowadays, master alloys and modifiers are one of the most promising areas for using granules 5–20 mm in size. An advantage of granules in comparison with master alloys in the form of ingots or bars is the fine microstructure of the material. This provides a large number of crystallization centers while modifying the alloys and providing a heredity of structure as well as the simplicity of batching and introduction of granules into the melt.

There are numerous application fields of aluminum granules in the areas of powder metallurgy due to satisfactory properties of the products produced from low-grade aluminum and the unique properties of highly alloyed granulated alloys. The basic kinds of aluminum granules, which are produced commercially, are given in [Table 15.5](#).

Impulse atomization is capable of producing droplets of controlled size having a relatively narrow distribution and a predictable cooling rate [13]. The scheme of the process, shown in [Fig. 15.7](#), consists of an impulse generator, a mechanical driver, a tundish, and a nozzle plate with orifices. The impulses generate monodisperse droplets that free fall out of the orifices. The quiescent gas atmosphere, into which melts have been atomized, may be helium, nitrogen, argon, or air. By varying the plunger acceleration, orifice size, orifice shape, melt temperature, and gas atmosphere, the granule size and size distribution, shape, and microstructure can be manipulated to the required specifications. Using this method, aluminum monodisperse granules from 2 to 10 mm in size are produced.

PRODUCTION OF FLAKES, PIGMENTS, AND PASTES

Aluminum flakes, pastes, or powders from fragile alloys, such as Al–Mg, Al–Ni, Al–Si, and others are produced by this method. For producing flakes, aluminum stock material (powder, foil) is crushed in a protective gas atmosphere (dry scheme) or in a liquid (wet scheme).

TABLE 15.5 Aluminum Granules and Pellets

Brand	Sieve Residue (wt%) (max) on Sieve With Aperture (mm)			Chemical Composition (wt%)									Applications	Source
				Al (wt%), min	Admixtures (wt%), max									
	7	15	20		Mg	Si	Fe	Mn	Cu	Zn	Ni	Sn		
AGHF	95.0	0.3		99.0		0.03	0.01		0.005		0.006		For cementation of gallium from aluminum solutions, as a chemical reagent etc	RUSAL
AL-110	0.297–0.841 mm (20–50 mesh)			99.7									Alloys, industrial-general	AEE
AG	95.0	5.0	0.3	96.0	1.0	0.6	3.0	0.5	0.3	0.5		0.3	In the metallurgical industry for deoxidization, modification of cast iron and steel	RUSAL
AL-130, pellets	6.35–12.7 mm (1/4" – 1/2")			99.99									Alloys, industrial-general	AEE
Granule 4121	0.25–2.0; max 5 wt% >2.0mm; max 2 wt % <0.25 mm												Shot peening, shot blasting	SA

RUSAL, Russian Federation Aluminum Company; AEE, Atlantic Equipment Engineers, Division of Micron Metals, Inc.; SA, produced by Sun Ark Aluminum Industries.

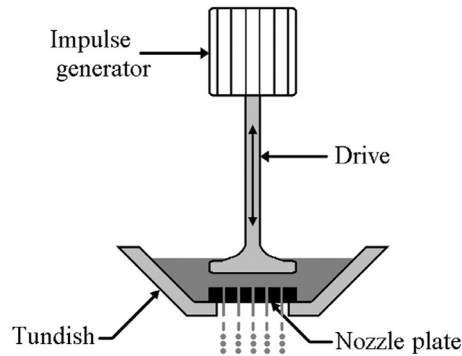


FIG. 15.7 Schematic of the impulse atomization process.

The basis of the technology for producing aluminum fine is the process of plastic deformation that provides the manufacture of particles in the form of thin flakes or leaflets. To prevent aluminum particles welding together and to reduce the friction between them and the milling balls, thus reducing the work required for deformation, stearin or another surfactant, in amounts from 0.3% to 4.0%, are added to the charge. Dry milling is carried out in a leak-tight mill working in a closed cycle in an atmosphere of nitrogen with the content of oxygen from 3% to 8%; the latter is necessary for the formation of a protective oxide film on the particle surface. At an oxygen content higher than 8%, an ignition inside the milling system is possible, and at an oxygen content lower than 2%, self-ignition of the discharged powder is possible.

The scheme for the production of pigment (PAP marks), pyrotechnic (PP marks) flakes, and lumpy flakes for sintering (AΠC marks, in Russian classification) is shown in Fig. 15.8. The main characteristics of aluminum flakes and lumpy aluminum flakes are presented in Table 15.6. Particles of aluminum lumpy flake are uniformly oxidized from the surface in the grinding process and agglomerated to the apparent density 0.9 g/cm^3 minimum. According to Russian GOST 10096–76, AΠC marks contain 6%–17% Al_2O_3 [14].

The process of AΠC lumpy flake production consists of two stages. In the first stage, aluminum stock material is ground in the presence of 0.5–1.2 wt% of fats. The flakes produced have a lamellar particle shape no more than $80 \mu\text{m}$ in size; the apparent density does not exceed 0.2 g/cm^3 . On decreasing the content of the fatty additive to 0.5%, a further oxidation (second stage) and cold welding of the aluminum particles takes place. Conglomerates (lumps) of fine particles with apparent density $1.0\text{--}1.4 \text{ g/cm}^3$ are formed. The sizes of particles aggregated during balling are basically of $300\text{--}500 \mu\text{m}$ but can achieve $1000 \mu\text{m}$.

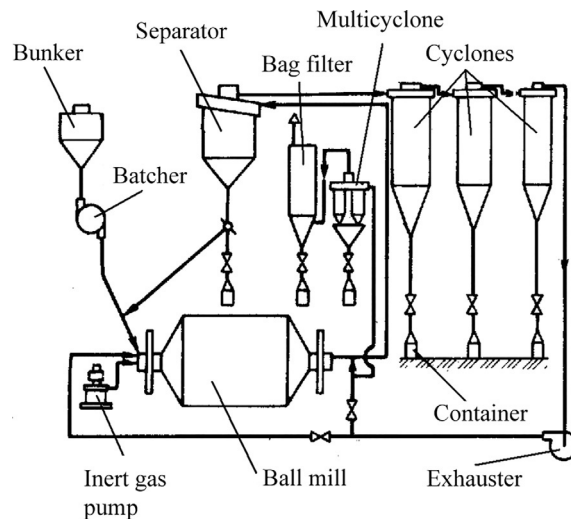


FIG. 15.8 Schematic diagram of the pneumoseparation grinding installation.

TABLE 15.6 Aluminum Flakes and Lumpy Flakes

Trade Name	Particle Size (μm) (max)	Covering Ability for Water Surface (cm ² /g) (min)	Apparent Density, (g/cm ³) (min)	Composition, wt%								Applications
				Al	Al ₂ O ₃	Impurities, wt% (max)					Fat Additives (max)	
						Mg	Si	Fe	Cu	Zn		
Aluminum flakes												
PAP-1 ^a	80	700	0.3	90	...	0.01	0.4	0.5	0.05	...	3.8	Production of paints, enamels, resin glues. As a gas developing agent to produce aerocrete is used in aircraft building, machine-building, instrument-making, polygraphic branch of industry. Production of explosives and pyrotechnic components
PAP-2 ^a	45	1000	0.2	85	...	0.01	0.4	0.5	0.05	...	3.8	
PP-1	160	...	0.3	97	...	...	0.4	0.5	0.08(Cu+Zn)		0.8	
PP-2	100	...	0.4	95	...	...	0.4	0.5	0.08(Cu+Zn)		0.8	
PP-3	63	...	0.5	93	...	...	0.4	0.5	0.08(Cu+Zn)		0.8	
TA 3700 G ^b	62 ^c	8000	0.17	96								Autoclaved aerated concrete (AAC), light weight concrete (LWC). Feedstock for chemical reactions, fuel cells and batteries, exothermic applications, binder agent for master alloy briquetting, drain cleaners, thermal and electrical conductivity enhancement in plastics, granular alternative for highly reflective roofing applications
K-109 ^d	600 × 600 × 25 (length × width × thickness)		0.47	99								
Aluminum lumpy flakes												
АПC-1A	315	...	0.9	Balance	6–8	...	0.25	0.20	...		...	Production of elevated temperature material SAP-1, capillary pipes and foil with the thickness from 0.3 to 0.1 mm resisting at 350–500°C. Production of profiled semiproduct. Production of semiproduct from SAP-2 with heat resistance higher by 15%–20% than SAP-1. Production of the semiproduct from the most heat-resisting material SAP-3
АПC-16	1000	...	1.0	Balance	6–8	...	0.25	0.25	...		...	
АПC-2	1000	...	1.0	Balance	9–12	...	0.25	0.30	...		...	
АПC-3	1000	...	1.0	Balance	13–17	...	0.25	0.30	...		...	

^aPAP-1 and PAP-2 produced for varnish-and-paint industry have the leafing of 80% min.

^bSwedish Carlfors Bruk Company.

^cδ₅₀—median diameter.

^dTransmet Corporation.

The charge of the fatty additive is increased with the increase of Al_2O_3 content in the final product:

(Al_2O_3) , wt%	6–11	11–17
Charge of the additive, wt%	0.6	0.6–0.9

Pastes are manufactured with petroleum-based or aqueous liquid phases. With the former, the aluminum flake content is of the order of 65% with 33% of volatile liquids such as white spirit, and 2% of fatty material (stearin acid). By the degree of fineness at grinding and the area of water coating ($1.5\text{--}3.0\text{m}^2/\text{g}$) on a mass unit, such paste considerably surpasses PAP-2—the finest aluminum flakes produced by dry grinding. Wet grinding is less explosive than the dry process of aluminum flake production.

The technological scheme that provides an opportunity of simultaneous or consecutive production of aluminum powders, flakes, and pastes includes the steps of initial material preparation, grinding, thickening, homogenizing, and casing (Fig. 15.9).

For the production of paste, aluminum powder manufactured by atomization, or scrap foil $<50\text{ }\mu\text{m}$ in thickness, liquid hydrocarbons (white spirit, xylene, toluene, etc.) and fatty additives (stearin and oleic acids, etc.) are used. Mixed material is ground in ball mills in one or several stages depending on the size of the initial aluminum particles and the required degree of particle fineness in the manufactured paste. The slurry from the mills is separated in a filter press, a drum-type filter, or in a centrifuge.

Aluminum pastes are more convenient in use and in transportation because of the absence of dusting, greater bulk weight, etc. Mixing the paste with a lacquer gives a paint, which has high moisture resistance and durability, high reflectivity, and heat resistance. Leafing and nonleafing aluminum pastes are produced. Aluminum particles of the leafing paste rise to the surface of the varnish film and arrange themselves parallel to the surface, which imparts a metallic shine and high reflectivity, complicates the penetration of moisture, and increases the stability of the coating. Aluminum particles of nonleafing paste settle in the bottom part of the varnish film, providing a polychromatic effect. Roof coatings comprise the largest market for nonleafing aluminum paste. The largest market for leafing aluminum paste is for automotive coatings.

Special application fields of aluminum paste are polygraphy, powder metallurgy, and the production of aerocrete.

Firms produce a multitude of aluminum pastes different in floating, granulometric composition, solvent, and special additives. Characteristics of the aluminum pastes produced in Russia are presented in Table 15.7.

There are two processes for aqueous aluminum paste. In the first process, aluminum flakes are mixed with an aqueous solution containing phosphate ions, glycols aryl alkyl sulfonates, carbohydrates, polyatomic spirits, etc., as an inhibitor. Such technology is mastered both at factories producing aluminum flakes and at some factories producing aerocrete. In

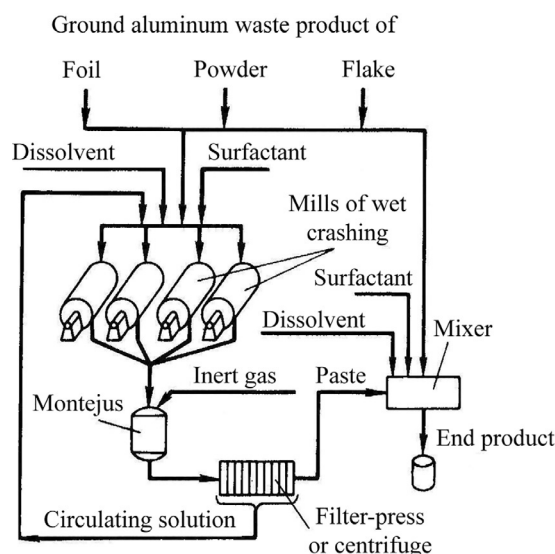


FIG. 15.9 Schematic diagram of manufacture of the aluminum pastes.

TABLE 15.7 Aluminum Pastes

Trade Name	Particle Size (μm) (max)	Covering Capacity on Water Surface (m^2/g) (min)	Leafing (wt%) (min)	Substances Soluble in Organic Solvents (wt%) (max)	Impurities (Fe, Si, Cu, Zn) total (wt%) (max)	Al (wt%)	Application
<i>Leafing aluminum pastes</i>							
B-1	80	1.6	65	3.0	1.0	Balance	Finish paints with metallic shine
B-2	45	2.0	75	4.0	1.2		
B-3	45	2.4	73	4.0	1.2		
<i>Nonleafing aluminum pastes</i>							
H-1	80	—	—	3.0	1.2	Balance	
H-2	45	—	—	4.0	1.2		
H-3	45	—	—	4.0	1.2		Priming coat protective paints
<i>Special aluminum pastes</i>							
C-1	80	0.7	65	3.0	1.2	Balance	Hammer enamels
C-2	45	1.7	75	3.5	1.2		
C-3	45	1.7	—	4.0	1.3		
MAII	160	1.3–1.7	50	3.5	1.0		Thermoregulating coatings
IID	80	2.0	30.0	3.5	1.0		Slushing paints and enamels
IIIDA	80	2.0	30.0	3.5	1.0		Paints and enamels in aerosol packing
GLS-70 ^a	$\leq 70\mu\text{m}$ ($\geq 95\%$)					80 minimum	Autoclave aerated concrete (AAC)
Silver lucule 9008 ^b	$\sim 45\mu\text{m}$	2.5	75	2–4		Balance	Exterior coatings, heat-resistant paint, anticorrosive

^aProduced by Xinji Guangyuan Metal Powder Co., Ltd.;^bProduced by Sun Ark Aluminum Industries (P) Ltd.

the second process, foil waste is ground in an aqueous solution containing fatty acids and inhibitors (salts of phosphoric and boric acids and colloid substances: albumin, casein, gelatin, glue, or arabic gum). Grinding is carried out at the lowest possible temperature of 283–323 K.

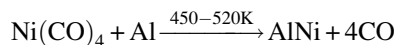
Water aluminum pastes containing 50%–65% of aluminum particles are fire- and explosion-proof, convenient to use, and the content of the active metal keeps the properties stable for a long time. They are used as a gas-forming agent to produce aerocrete, in the production of paints, in the structure of water-filled explosives, and in powder metallurgy.

The same technique is used for producing a water paste containing 20 wt% Al_2O_3 . This paste is thickened, dried, and ground to a powder state. Thus, a “heavy” powder with the apparent density above 1.2 g/cm^3 suitable for use in PM is manufactured.

NICKEL-CLAD ALUMINUM POWDERS

A special class of aluminum powder materials is composite materials, which include powders of aluminum and its alloys or mixtures of such powders with powders of other compounds clad by nickel.

The production technology of the specified composite materials developed by the Norilsk Nickel Concern consists of coating powder particles of aluminum and its alloys with metallic nickel by the carbonyl process, that is, by the decomposition of nickel tetra carbonyl vapor on the surface of the powder suspended in the vibrofluidized bed:



Properties of Nickel-Clad Aluminum Powders

- Powder composition, wt%: 4–23 Al; 96–77 Ni.
- Powder properties
 - Thickness of the nickel coating on a particle of aluminum: from one to tens of microns.
 - Shape of particles: close to spherical.
 - Flow rate: 10–40 s/50 g.
 - Apparent density: 2.5–5.1 g/cm³ (depending on the content of nickel).
 - The size of particles: 50–100 μm.
- Properties of plasma coatings:
 - Microhardness: 4.0–8.0 GPa.
 - Porosity: 3%–10%.
 - Strength of adhesion to the substrate: 30–38 MPa.
 - Exothermal effect while spraying of coatings comes at 898 K due to the reaction heat of the synthesis of the nickel-aluminum intermetallic in isothermal conditions.
 - Thickness of the coating layer that excludes peeling from the substrate: on Steel-3: 1.0 mm; on Steel-45: 0.8 mm; on D16T alloy: 3.0 mm (The steel brands are given in Russian classification).
- Purpose: spraying of an intermediate layer on steels, cast irons, aluminum, and titanium alloys, etc.
- The addition data about nickel-clad powders are contained in [Chapter 12](#).

ALUMINUM SCRAP RECYCLING

Waste Aluminum Foil Recycling

The great significance of recycling is for environmental and economic aspects of saving material and energy. Especially, the recycling of aluminum scrap is important because the production of aluminum requires a lot of energy. Almost 100% of the use of foil waste products is achieved by their processing to aluminum powders (grit), flakes, or paste.

The technology of foil processing is quite simple: crushing in cutting units such as the ones used for comminution of paper and plastic with subsequent use as the raw material for the production of aluminum flakes (dry crushing) or pastes (wet crushing). After grinding in mills (ball, jet-type, etc.), disintegrators, and the like, the crushed flat particles of foil become rounded in shape, forming a special kind of powder product named “grit.” Grit has a wide application in the chemical industry as a reducing agent or gasifier, and in other industries as well.

Aluminum flake powders have been widely used in fingerprint detection, aerated concrete, as a paint component for automobile and industrial applications, an additive for ink, and explosives due to their metallic color, glossy appearance, and good adherence. The aluminum flake powders of mean size 20–45 μm and water covering area of 4000–30,000 cm²/g are mainly used in the industry. In the paste manufacturing industry for paints and inks, flake powders are produced by the wet ball-milling process of gas-atomized aluminum powder. The wet ball milling is performed in an inorganic solvent such as mineral spirits.

A study for producing aluminum flake powder from aluminum foil scrap by dry ball milling under a gaseous argon with a small percentage of oxygen was carried out [15]. Aluminum foil scrap of thickness 6.5 mm, width 8 mm, and purity 99.4 wt% was used as the initial milling material. The mill drum was filled with argon gas containing 8 vol% oxygen, which slightly reacts with the surface of the aluminum powder, preventing the explosion of the milled fine powder after exposure

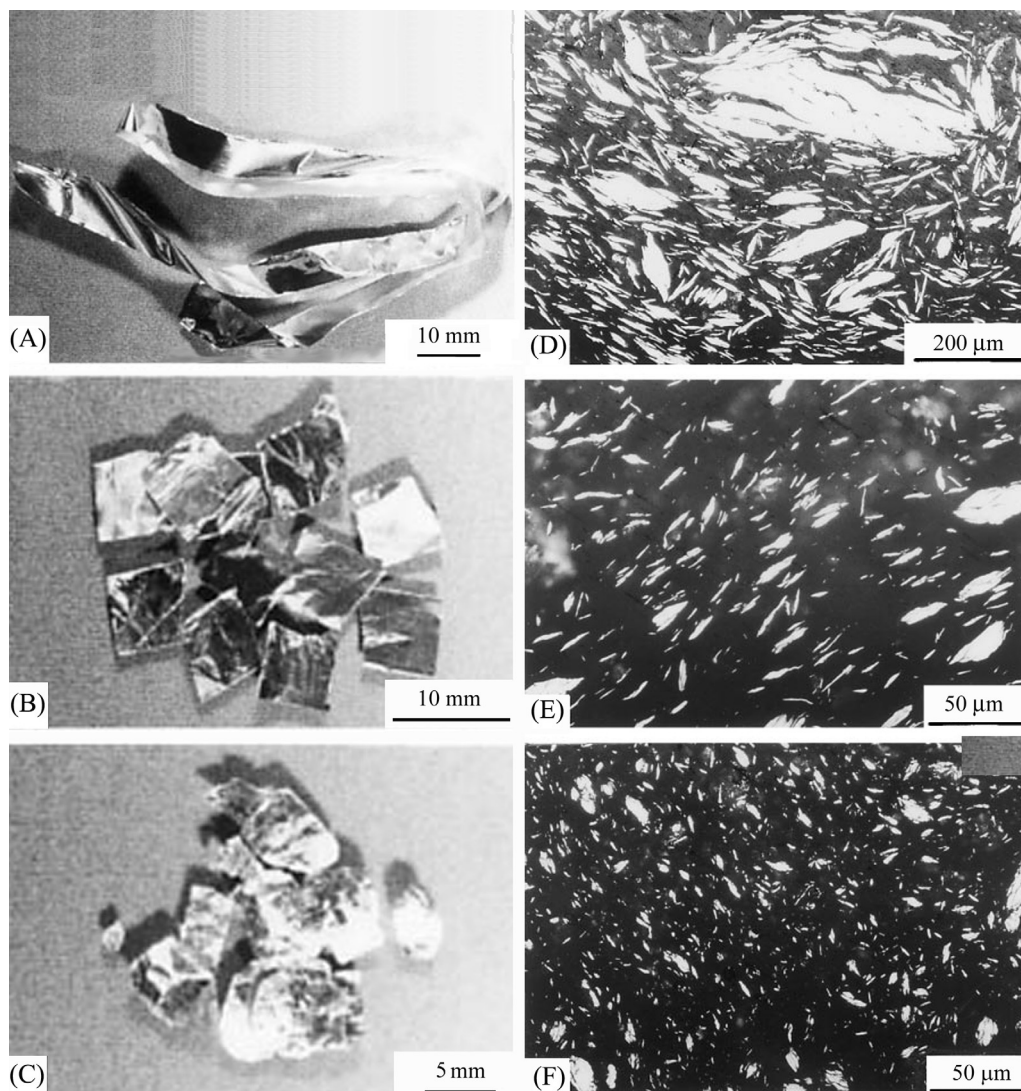


FIG. 15.10 Optical micrographs of aluminum flake powder during making from foil scrap for different milling times: (A) foil scrap with a thickness of 0.65 mm; (B) cut foil of 6×7 mm; (C) foil ball milled for 5 h; (D) flake powder ball milled for 10 h; (E) 20 h, and (F) 25 h. (From Hong SH, Lee DW, Kim BK. Manufacturing of aluminum flake powder from foil scrap by dry ball milling process. *J Mater Process Technol* 2000;100:105–9.)

in air. The content of stearic acid was varied from 1.5 to 5 wt% in order to check the effect of the level of stearic acid. The milling was performed for 45 h at a rotational speed of 120 rpm.

Optical micrographs of aluminum flake powder during manufacture from foil scrap for different milling times are presented in Fig. 15.10. During the milling of the cut foil ($6.5 \times 6 \times 8$ mm) using steel balls of diameter 16 mm for 5 h, the foil was microforged by the falling of the balls and deformed plastically into an elongated shape. Most of the aluminum foil is transformed into coarse flake powder by milling for up to 10 h (Fig. 15.10D). As shown in Fig. 15.10E, coarse flake powder is transformed into smaller flake powder with a single layer, or a laminated form occurs with crack nucleation and growth in the laminated flake powder after milling for 15 or 20 h. As the milling time increases up to 25 h, the powders are work-hardened and fragmented into fine flake powder, as shown in Fig. 15.10F.

A larger amount of stearic acid reduces friction between the foil and the balls and vial wall, thus giving less milling efficiency and a 3 wt% of stearic acid as the additive being verified as the optimum content. The variation of the mean powder size with the cooling of the milling material by an intermediate stop during the milling of aluminum foil for 25 h was examined. The powder milled continuously for 25 h shows 62 μm mean size, but the powder, which was milled with three intermediate stops (10, 15, and 20 h, respectively) during milling for 25 h, has a mean size of 14 μm . This result was interpreted by the elevated temperature behavior of the prolonged continuous milling process, thus the milled

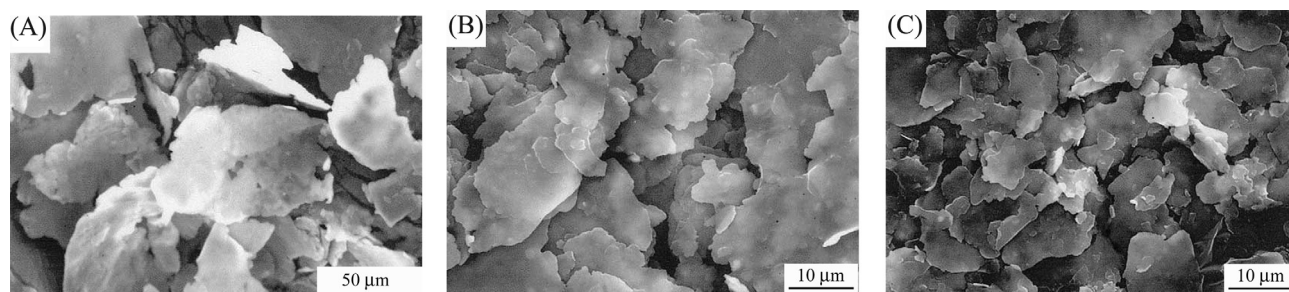


FIG. 15.11 SEM micrographs of aluminum flake powder wet milled for 30h with foils having different initial thickness: (A) 60 μm , (B) 16 μm , and (C) 6.5 μm . (Modified from: Hong SH, Kim BK. Fabrication of aluminum flake powder from foil scrap by a wet ball milling process. *Materials Letters* 2001; 51 (Issue 2, October): 139–43.)

aluminum powder became more ductile, leading to the agglomeration of incipient particles by welding and autoadhesion. The intermediate stops lead to decreased temperature and less ductility. However, such techniques decrease the powder output and increase the power consumption.

The aluminum flake powder produced by the dry ball milling of aluminum foil scrap can be used in fingerprint detection or to aerate concrete.

Wet ball milling is used mainly in the paste manufacturing industry for paints and inks, in which milling is fulfilled in an inorganic solvent. The fabrication of aluminum flake powder by the wet ball milling of aluminum foil scraps and the effect of related factors were studied in the work [16]. The aluminum foil scraps of various thickness (6.5, 16, 60, and 120 μm), 8 mm in width, and a purity of 99.4 wt% were used as the initial milling material. The milling was performed for 60 h under a rotation speed of 120 rpm. In the capacity of inorganic solvent, the mineral spirits was used. The main variable parameters in the wet ball-milling process were the initial foil thickness 6.5, 16, 60, and 120 μm , the content of oleic acid in the 0%–5% range, the weight ratio of mineral spirits to foil in the 0.5–3.0 range, and a ball sizes of 16 and 4.8 mm. Oleic acid of 2–3 wt% as the additive and a weight ratio of 1.0–3.0 of mineral spirits to foil were evaluated as the range of optimum content. Fig. 15.11 shows aluminum flake powders prepared by milling of foil scraps with different initial foil thickness for 30 h. In comparison to the foil scrap with an initial thickness 60 mm, the one with 6.5 mm in thickness was milled to a finer flake powder.

Hong et al. reported [16] that the water-covering area of flake powder is increased with milling time due to the increase of the specific surface area of milled powder. Flake thickness of powder is decreased continuously by milling up to 30 h. For the foil of 6.5 mm in thickness, as milling time increases to 30 h, the aspect ratio of flake powder is increased to the maximum due to a high degree of flake. After milling for 60 h, flake powder is fragmented into fine powder with a low degree of flake. As the content of oleic acid increases from 1.5% to 5%, the mean size of powder milled for 30 h is decreased drastically due to the prevention of coagulation, as shown in Fig. 15.12.

For the mineral spirit content below 50%, foil scrap with an initial thickness of 6.5 μm was not milled because of the sliding motion of balls coated by oleic acid. As the weight ratio of mineral spirit to foil increases over 100%, foils were milled to powders with a mean powder size of 15–20 μm . After milling for 30 h, the content of mineral spirits in aluminum paste was controlled at 35% below by vacuum drying for pigment application.

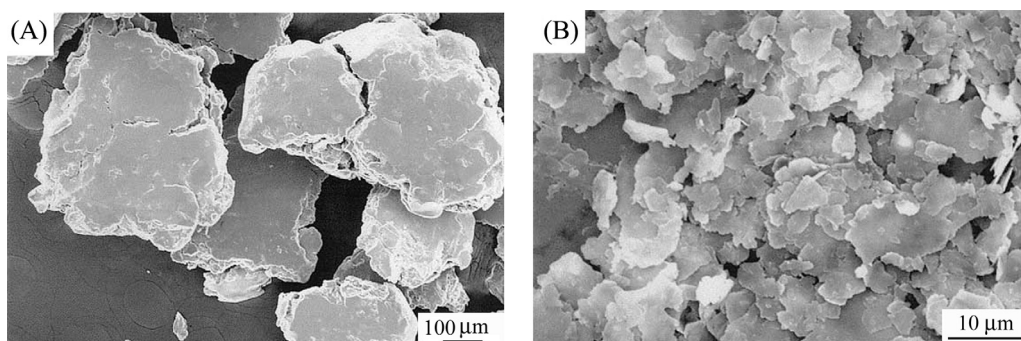


FIG. 15.12 SEM micrographs of aluminum flake powder wet milled for 30h: (A) without and (B) with 5 wt% oleic acid content.

Processing of Recycled Aluminum to Granules

In recent years, the processing of aluminum scrap (mainly chips and used drink cans) has attracted increasing attention. Aluminum scrap has a complex chemical composition based on aluminum (>90%). Magnesium, zinc, silicon, iron, etc., are the main impurities. Sources of impurities are the composition of the alloy for producing drink cans (mainly, Al–Mg alloy containing up to 3%–4% Mg) and mechanical impurities that enter into the scrap due to inefficient sorting, classification, and storage of the scrap.

Nowadays, all over the world, the main method of processing aluminum scrap is remelting in units of various types. The melt is usually cast into ingots. In some cases, the melt is granulated or gas atomized to produce powder of recycled aluminum (for example, APV powders in Russia). These remelting-casting technologies are rather power consuming.

Processing of aluminum scrap to powders and granules is an alternative to remelting scrap to produce ingots. The advantage of this technology is that it is a simpler and cheaper process of producing granules from recycled aluminum. The size of granules is easily regulated by the adjustment of grinders and varies from 1 to 10 mm. Such granules are used more effectively than ingots in ferrous metallurgy for steel deoxidation.

THE MAIN APPLICATION FIELDS FOR ALUMINUM POWDERS

Powders and flakes of aluminum are ingredients of many industrial explosives used in the mining industry (ammonal, rock ammonites). As a high-energy carrier, aluminum essentially raises the heat of the explosive transformation and the efficiency of the blasting operations. Powders such as PA, pyrotechnic flakes of PP marks, powders from secondary aluminum such as APV, PA-VV, etc., (in CIS classification) are the most widely used.

The aluminum is used to increase the energy and raise the flame temperature in rocket propellants. Aluminized explosives have been used in various formulations since the beginning of the 20th century. It is incorporated in explosives to enhance air blast, increase bubble energies in underwater weapons, raise reaction temperatures, and create incendiary effects [17–19]. Nanometric aluminum powder has recently become available in quantities large enough for introduction into energetic materials and small-scale performance testing. The most widely known product is the ultrafine aluminum powder “Alex,” which comprises spherical particles of 100–200 nm size.

Various plastic-bonded explosives (PBXs) and trinitrotoluene (TNT)-based formulations have been developed to compare ultrafine and conventional micron-sized aluminum [20]. Explosive performance was determined by velocity of detonation (VOD) measurements and plate dent depth tests. The study was also complemented by air-blast tests to evaluate the difference in energy release in the far field. Brousseau et al. reported [20] that it was found that for PBX compositions, no significant differences were observed between formulations containing micron-sized and nanometric aluminum. For mixes of TNT and aluminum, an improvement was noted in the velocity of detonation and in plate dent depths using ultrafine aluminum. Aquarium tests were carried out on TNT/Al blends to confirm the improvement observed, to evaluate the relative performance of the different sizes of aluminum, and to qualify the behavior of the aluminum in such an explosive. No increase of pressure was found in aquarium tests. However, the expansion of the gases appears faster for TNT-type formulations containing nanometric aluminum.

In the study of the acceleration ability of high explosives by aluminum nanopowder additives, Davydov and Gubin [21] used two types of powders. The first was obtained by exploding wires in a nitrogen atmosphere. The particle size ranged from 50 to 300 nm. The content of active Al was approximately 89%. The second powder, made by the plasmochemical method, was passivated with a 2% silicone coating. This powder has an amorphous and, hence, more permeable oxide film and possesses an extremely high chemical activity. The content of active Al was approximately 87%. The study, similarly to the work [20], showed that the introduction of various Al additives into high explosives reduces the detonation pressure while raising the temperature of the detonation products. The authors [21] explained this phenomenon by the following main features of the behavior of metallic particles in the composition of high explosives during their detonation:

- The difference in the shock compressibility of the high explosives and the metal additive should probably lead to the creation of zones of increased pressure due to shock wave reflection from the metal particles, which manifests itself in the known sensitizing effect of metal additives.
- During shock wave loading, due to differences in the physical and mechanical properties of Al and Al oxide, the destruction of the latter occurs.
- Due to differences in the shock compressibility of high explosives material, Al, and its oxide, a high explosives decomposing material behind the detonation complex shock front flows around Al particles. As a result, the oxide peels off the surface of Al particles, thereby ensuring the supply of oxidizer to the particle surface and the removal of the reaction products.

Two compositions of aluminized explosives were developed for an investigation of the effects of boron powder on the air blast properties [22]. These had the following mass compositions: 35% of metal powders, 20% of ammonium perchlorate (AP), 36% of hexogen (RDX), and 9% of wax ($(CH_2)_n$) and other additives (including 1% graphite). Charges of 500 g of the aluminized explosives were prepared and studied. Their trinitrotoluene (TNT) equivalences were calculated according to the experimental data and the explosion law.

The authors reported [22] that the highest surface temperatures of the fireballs of these aluminized explosives were up to 1870 K, which was higher than that of the TNT charge. In the region of the highest surface temperature above 970 K, the duration for the composition RDX/Al/AP was about 232 ms (2.73 times more than TNT) while RDX/Al/B/AP (by 25 wt% Al/10 wt% B ratio) was about 360 ms. The fireballs obtained from the explosion of these aluminized explosives had larger dimensions than that of TNT. Therefore, for the two compositions, RDX/Al/B/AP had a higher surface temperature, a longer duration of the explosion fireball, and when the surface temperature was above 1273 K, it had larger fireball dimensions than the other composition. The authors [22,23] considered that aluminum and boron powders act as inert additives in the reaction zone. The secondary oxidation reactions of aluminum and boron with the expanding detonation products contribute to late-time effects, such as the air blast shock wave, postdetonation fireballs, and detonation temperature.

In ferrous metallurgy, aluminum powders (granules) are used as a reducing agent. Introduction of 0.1%–1.5% of aluminum groats into the melt of iron completely releases it from the oxide (II) and thereby provides the production of a dense casting without blowholes. Aluminum powders are also used for heating the riser part of the ingot in the casting of ferrous metals. A mixture of oxides and aluminum powders, tamped around the sprue opening and the riser, which “ignites” from the melt and during burning, provides cast metal in the liquid state for a long time. This allows, in the time of the melt solidification following by decreasing of its volume, adding melt to the mold and avoiding the formation of shrinkage cavities in casings.

Aluminothermic reduction of refractory metal oxides is used in the production of ferromolybdenum, ferroniobium, ferrotungsten, etc.

The thermite mix containing 75% Fe_2O_3 and 25% Al provides a burning temperature up to 2773 K.

In the chemical industry, aluminum powders and granules are active reducers as they rapidly react in aqueous solutions of acids or alkaline media with the emission of hydrogen: 1.0 kg of aluminum powders gives 1240 L of hydrogen. They are applied in the production of aluminum alkyls, polymerization of ethyls, for production of antidetonants, tetraethyl lead, etc.

Aluminum powders are used for coating steel products by the methods of spray deposition, flame or plasma spraying, and thermal diffusion. Aluminum powders and pastes are used as a paint and protective pigment in compositions of various paints, varnishes, and enamels. The paints obtained have a beautiful silvery color while protecting products from moisture, heat, and light.

The American Society for Testing and Materials (ASTM) has enacted specifications for aluminum flake pigments (ASTM D 962, Part 28) [24]. The standard lining paste (ASTM D 962, type 2 class A) is used for structural steel coatings.

Aluminum-based sintered materials are distinguished in good pressing and sintering ability of powders, manufacturing efficiency, high strength, and good corrosion resistance.

In Russia, a range of Al-based powders is produced such as lumpy powder АПЦ, powders of high-purity aluminum (PA-VCh), of technical-purity aluminum such as PA, aluminum groats AKP, and powders of SAS alloys (Al–Si). These materials are destined for PM production in machine building and electrical engineering as well as in the nuclear, electronics, and other industries.

Porous aluminum shows the promise of heat sink cooling materials due to its unique combining of high heat conductivity with small mass density. In the case of using a boiling cooling system for heat removal from the electronic equipment, harnessing latent heat with refrigerant, the advantage of the system is that it does not require a water pump and extra power source, which are necessary equipment for a forced water-cooling system. For better performance of the ebullient cooling system, it is important to boil refrigerant efficiently on the surface of the boiling heat transfer. Recently, applying porous aluminum for the boiling heat transfer surface has been widely reported [25,26]. In the work [27], a three-dimensional structure with porous aluminum is discussed. The product making comprises the following major steps. First, pure aluminum powders ($\delta_{50} = 6 \mu\text{m}$, purity 99.7%) are mixed with a polyvinyl alcohol water solution and some reagents are added to make an aluminum powder slurry. Next, the slurry is coated on urethane form and dried by hot air. This green body was heat-treated by using a vacuum heat-treatment furnace and then sintered in a vacuum. Fig. 15.13 shows SEM images of the porous aluminum sample: in initial condition (left) and after etch and ultrasonic treatment (right). The latter shows the 0.5–2 μm size particles on the sintering surface. The result showed the boiling heat transfer coefficient of the etched and ultrasonic treated sample reached the highest value, 16 kW/($\text{m}^2 \cdot \text{K}$). In contrast, the boiling heat transfer coefficient of the ingot machined sample showed an even configuration amount of 7.5 kW/($\text{m}^2 \cdot \text{K}$).

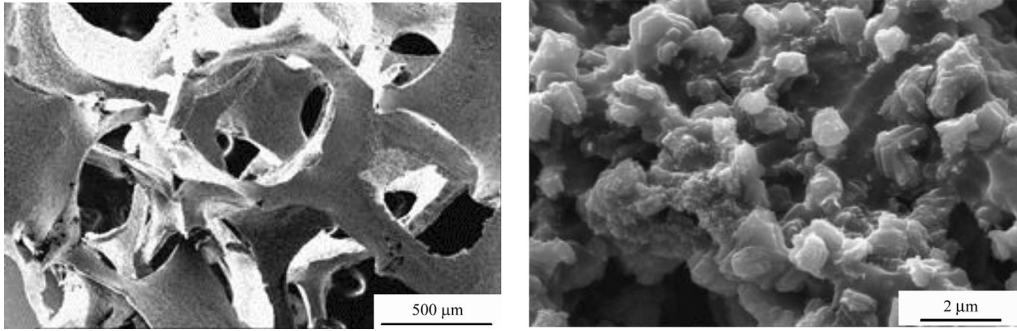


FIG. 15.13 SEM images of porous aluminum sample: in initial condition (*left*) and after etch and ultrasonic treatment (*right*).

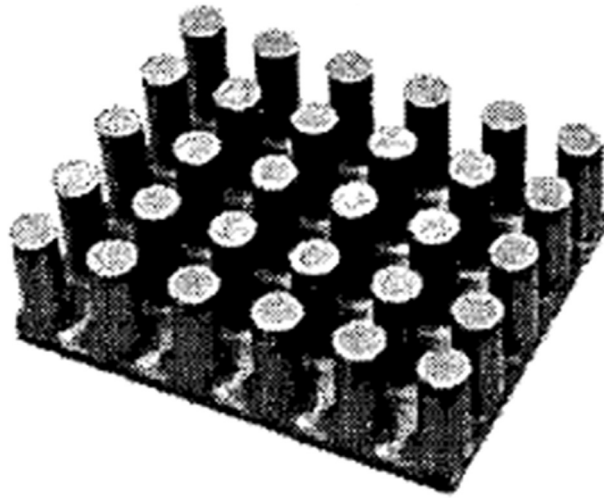


FIG. 15.14 Fin heat sink produced by powder injection molding (PIM) process.

Tan and Ma considered [28,29] the making of aluminum components by powder injection molding (PIM) for heat sink application. The authors reported [28] that the thermal conductivity of a fin heat sink (Fig. 15.14) produced by the PIM process in comparison with a machined square pin in the same setup is about 30% higher, accordingly $180 \text{ W/(m}\cdot\text{K)}$ and $135 \text{ W/(m}\cdot\text{K)}$.

Aluminum powders, flake pigments, pastes, paints, and granules are applied in:

- Production of high density refractory materials (to increase the thermal resistance and mechanical strength of the refractory material).
- Production of porous adsorption materials for cryogenic equipment.
- Production of a welding wire and cutting emulsions.

Aluminum powders are added to rubber to produce automobile tires with the purpose of increasing the heat conductivity of the protective rubber and to improve the operating conditions for the inside tire surface. An addition of powders to plastics improves their mechanical properties, increases their wear resistance, and imparts special properties (electric conductivity, heat conductivity, etc.).

Aluminum powder is applied as a filler material in plastics such as epoxy, polyester, and phenolic resins. In particular, powder aluminum is used in the preparation of so-called “cold solders” (fillers), which are used for filling of hollows, cracks, and seams on the surface of a metal. $>50\%$ of the content of the cold solder is aluminum powder, which is mixed with the resin and hardener. Aluminum powder is used as a lubricant in producing ceramic details by the method of dry pressing for the reduction of the wear in molds, the obtaining of high-density compacts, and the reduction of the pressing force.

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